```
!pip install -q -U langchain langchain-groq
```

```
from langchain.agents import create_agent
from langchain.tools import tool
from langchain_groq import ChatGroq

print("Libraries imported successfully!")
```

```
Libraries imported successfully!
```

```
from google.colab import userdata

GROQ_API_KEY = userdata.get("Secret_Key")

if GROQ_API_KEY:
    print("Groq API key loaded successfully!")
else:
    print("Groq API key not found!")
```

```
Groq API key loaded successfully!
```

```
llm = ChatGroq(
    model="llama-3.3-70b-versatile",
    temperature=0,
    api_key=GROQ_API_KEY
)

print("Groq LLM initialized successfully!")
```

```
Groq LLM initialized successfully!
```

```
@tool
def calculator(expression: str) -> str:
    """
    Calculate a mathematical expression.
    Use this tool whenever the user asks for arithmetic calculations.
    """

    try:
        result = eval(expression, {"__builtins__": {}}, {})
        return str(result)

    except Exception as e:
        return f"Unable to calculate: {e}"
```

```
@tool
def knowledge_base(query: str) -> str:
    """
    Provides basic information about common computer science and AI topics.
    Use this tool when the question requires factual information from the
    available knowledge base.
    """

    knowledge = {
        "python": "Python is a high-level, interpreted programming language widely used in AI, data science, automation and

        "artificial intelligence": "Artificial Intelligence is the field of creating systems that can perform tasks that no

        "machine learning": "Machine Learning is a subset of AI where systems learn patterns from data to make predictions

        "deep learning": "Deep Learning is a subset of machine learning that uses neural networks with multiple layers.",

        "react": "ReAct stands for Reasoning and Acting. An agent reasons about a problem, selects an action or tool, obser

        "langchain": "LangChain is a framework for developing applications and agents powered by language models."
    }

    query_lower = query.lower()

    for key, value in knowledge.items():
        if key in query_lower:
            return value

    return "No matching information was found in the knowledge base."
```

```
tools = [
    calculator,
    knowledge_base
]
```

```
agent = create_agent(
    model=llm,
    tools=tools,
    system_prompt="""
You are a helpful question-answering ReAct agent.

You should reason about the user's question and decide whether a tool
is required.

Available tools:
1. calculator - for mathematical calculations
2. knowledge_base - for basic AI and computer science information

Use a tool when it is useful.
After receiving the tool result, use that observation to formulate
the final answer.

Always provide a clear and concise final answer to the user.
"""
)

print("ReAct agent created successfully!")
```

```
ReAct agent created successfully!
```

```
question = input("Enter your question: ")

result = agent.invoke(
    {
        "messages": [
            {
                "role": "user",
                "content": question
            }
        ]
    }
)

print("\n========== FINAL ANSWER ==========")

final_message = result["messages"][-1]

print(final_message.content)
```

```
Enter your question: 25*16
```

```
========== FINAL ANSWER ==========
The result of 25*16 is 400.
```

```
print("========== REACT AGENT TRACE ==========\n")

for message in result["messages"]:

    print("MESSAGE TYPE:", type(message).__name__)

    if hasattr(message, "content") and message.content:
        print("CONTENT:")
        print(message.content)

    if hasattr(message, "tool_calls") and message.tool_calls:
        print("TOOL CALL:")
        print(message.tool_calls)

    print("-" * 70)
```

```
========== REACT AGENT TRACE ==========

MESSAGE TYPE: HumanMessage
CONTENT:
25*16
----------------------------------------------------------------------
MESSAGE TYPE: AIMessage
TOOL CALL:
[{'name': 'calculator', 'args': {'expression': '25*16'}, 'id': 'fs55qzttg', 'type': 'tool_call'}]
----------------------------------------------------------------------
MESSAGE TYPE: ToolMessage
CONTENT:
400
----------------------------------------------------------------------
MESSAGE TYPE: AIMessage
CONTENT:
The result of 25*16 is 400.
----------------------------------------------------------------------
```

```
question = "What is artificial intelligence?"

result = agent.invoke(
    {
        "messages": [
            {
                "role": "user",
                "content": question
            }
        ]
    }
)

print("========== FINAL ANSWER ==========")
print(result["messages"][-1].content)
```

```
========== FINAL ANSWER ==========
Artificial intelligence (AI) refers to the development of computer systems that can perform tasks that typically require hum
```

```
question = "What is ReAct in AI agents?"

result = agent.invoke(
    {
        "messages": [
            {
                "role": "user",
                "content": question
            }
        ]
    }
)

print("========== FINAL ANSWER ==========")
print(result["messages"][-1].content)
```

```
learn from their experiences. In the context of our conversation, I'm using the ReAct framework to understand your question,
```

```
print("========================================")
print("       SIMPLE ReAct Q&A AGENT")
print("========================================")
print("Type 'exit' to stop.\n")
```

```
while True:

    question = input("You: ")

    if question.lower() == "exit":
        print("Agent: Goodbye!")
        break

    result = agent.invoke(
        {
            "messages": [
                {
                    "role": "user",
                    "content": question
                }
            ]
        }
    )

    answer = result["messages"][-1].content

    print("\nAgent:", answer)
    print("-" * 60)
```

```
========================================
       SIMPLE ReAct Q&A AGENT
========================================
Type 'exit' to stop.


Agent: I'm not able to provide real-time information or current weather conditions. However, I can suggest some ways for you
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Agent: Agentic AI in sales force refers to the use of artificial intelligence (AI) systems that can perform tasks autonomous

Agentic AI in sales force can be used for a variety of tasks, such as:

1. Lead qualification: Agentic AI can analyze customer data and behavior to qualify leads and determine which ones are most
2. Sales forecasting: Agentic AI can analyze historical sales data and market trends to predict future sales performance.
3. Customer segmentation: Agentic AI can analyze customer data to identify patterns and segment customers based on their beh
4. Personalized recommendations: Agentic AI can analyze customer data and provide personalized product or service recommenda
5. Automated communication: Agentic AI can automate communication with customers, such as sending emails or chat messages, t

The use of agentic AI in sales force can bring several benefits, including:

1. Increased efficiency: Agentic AI can automate routine tasks, freeing up sales representatives to focus on high-value acti
2. Improved accuracy: Agentic AI can analyze large amounts of data and provide accurate predictions and recommendations.
3. Enhanced customer experience: Agentic AI can provide personalized recommendations and interactions, leading to a better c
4. Increased sales: Agentic AI can help sales representatives identify and pursue high-value opportunities, leading to incre

However, the use of agentic AI in sales force also raises several challenges and concerns, such as:

1. Data quality: Agentic AI requires high-quality data to function effectively, which can be a challenge in sales force wher
2. Bias: Agentic AI can perpetuate biases and discrimination if the data used to train it is biased.
3. Transparency: Agentic AI can be opaque, making it difficult to understand how decisions are made and actions are taken.
4. Job displacement: Agentic AI can automate tasks, potentially displacing sales representatives and other jobs.

Overall, agentic AI has the potential to transform the sales force by providing personalized recommendations, automating rou
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Agent: The term "exist" is a broad concept that can be related to various fields such as philosophy, mathematics, and scienc
```